EVERYDAY TOOLS HUB

Product requirements, market-gap analysis & launch plan

Version 2.0 — a refactor of v1.0. Real market gaps researched on the live web, and a new AI-era tool category (“AI Prep”) designed to fill them.

STATUS	Product direction approved (v1). **AI Prep category is proposed — needs your decision** (Section 14.3).
POSITIONING	Balanced global utility hub — unchanged, still a confirmed decision.
VISUAL DIRECTION	Playful, friendly, clear and trustworthy — unchanged.
DOMAIN	Undecided; naming and availability research still deferred.
VERSION	2.0	11 November 2026	supersedes everyday_tools_hub_prd_v1.docx

RESEARCH INTEGRITY v1 was written without web access, so every competitor and demand claim was unverified. For v2, **12 live web searches were run on 11 November 2026.** Almost all results are secondary sources — blogs, news, vendor pages — and several are written by competitors selling a rival product.

Every finding therefore carries a confidence rating (Section 3.1). Treat the findings as **well-sourced hypotheses, not facts**. They do not replace the hands-on tests, keyword data and primary-source checks in Section 12.

Document guide

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1. What changed in v2

v1 is a solid, honest specification for twelve everyday tools. Its weakness is strategic, not editorial: it planned to compete head-on with incumbent tool sites using a differentiator (“trust”) it had not yet evidenced, and it ruled out every AI-adjacent need. v2 keeps everything that was sound and fixes the following.

Issue found in v1	Why it matters	Fix in v2
Differentiation rested on generic “trust” and “privacy” claims, with no evidence of an underserved need.	Twelve commodity tools versus established brands is a hard fight for a new site with no authority.	Section 3 maps ten evidenced problems to specific solutions. A new AI Prep category targets needs the incumbents do not obviously serve.
Non-goals excluded “AI-powered rewriting, OCR, image generation or document understanding” as one block.	Lumped together generative AI (which we should avoid) with preparing and checking data for AI (which users clearly need).	Non-goals narrowed: no generative AI, no server-side inference. On-device, task-specific models are allowed behind strict gates (§7.8).
No treatment of AI search (AI Overviews, zero-click) on traffic.	The entire acquisition and AdSense plan assumed classic organic clicks.	Section 12.4 adds a distribution plan for the AI-search era and treats traffic as volatile; Section 13.4 adds a monetization fallback.
Single revenue path (AdSense) with no contingency.	One policy decision or traffic shock removes all income.	Decision D6 opens a diversification review; AdSense remains the first path.
Competitor register and keyword tables were empty placeholders.	Cannot prioritize without evidence.	Register seeded with 12 dated, rated findings (§3.2). Still needs hands-on verification.
Research templates repeated across Sections 11, 12 and Appendix A.	Long, duplicated, hard to maintain.	Consolidated into one research plan (§12) and one template appendix.
Tool registry had no field for processing location, accuracy or model assets.	Cannot enforce “don’t claim local until audited” in code.	Registry v2 (§9.3) makes the privacy claim and accuracy label data-driven and audit-gated.
Tools were isolated: no shared text/data pipeline and weak cross-linking.	Misses the “family of tools” benefit the thesis relies on.	Shared text workspace and detector engine (§9.2); explicit cross-tool links in §6.4.

Kept unchanged: the positioning, visual direction, domain deferral, English-first launch, client-side-first processing, the tool-first page structure, the P0 definitions, the opportunity rubric and the AdSense readiness gates.

2. Executive summary

Everyday Tools Hub is a free, English-first website that helps people finish common digital tasks quickly — compress images, merge PDFs, count words, format JSON, make QR codes, calculate dates — without accounts. v2 adds a seventh category, **AI Prep**: tools that help people get data safely and correctly into AI systems, and check what comes back out.

Product statement

> **POSITIONING** Useful online tools, ready when you need them — and now ready for the AI era too. Clear controls, honest results, processing you can verify, and no account.

The three big moves

#	Move	What it means
1	**Gap analysis**	Ten problems found in the current tool landscape, each with evidence, a confidence rating and a proposed solution (§3).
2	**AI Prep category**	Four Wave-1 tools (LLM JSON Repair, Safe-to-Paste Redactor, Token & Context Calculator, Hidden Text Scanner), two Wave-2 tools (Doc→LLM Markdown, Image Metadata Inspector) and three gated on-device AI tools (§6, §7).
3	**Provable trust**	Local processing becomes a verified, dated claim shown on the page, not a slogan. Every estimate is labelled Exact, Estimated or Heuristic. Every automatic change is shown as a diff (§6.4).

Product thesis (revised)

- People search for task-specific solutions, not directories. Each tool keeps a focused landing page answering one intent. (Unchanged from v1.)
- The **commodity** tools (compress, merge, convert) are the traffic foundation but not the moat: incumbents are large and mostly upload files to servers. Browser-local processing is a credible but **unproven** differentiator.
- The **AI Prep** tools target problems that are new, concrete and under-served by the incumbents' core product lines — sensitive data pasted into chatbots, broken model output, documents that LLMs cannot read well, hidden text in files. Demand is evidenced by behaviour and research, but **search volume is unmeasured**.
- Traffic and AdSense revenue remain outcomes to validate. Search is shifting under AI Overviews; the plan must survive a smaller click pool.

> **BOTTOM-LINE RECOMMENDATION** Approve the AI Prep category as a **measured experiment inside the balanced hub**, not a pivot. Build the 12 core tools to the original standard, add the four cheap, text-only Wave-1 AI Prep tools alongside them (they share one workspace and need no heavy models), and let Search Console data decide whether Waves 2 and 3 are built.

3. Market gap analysis

This section answers: what is missing in the tools people can use today, what evidence is there, and what would we build to fix it?

3.1 Method and confidence scale

Twelve searches covered: privacy of upload-based PDF tools, PDF→Markdown for LLMs, employee data leakage into AI, token counters, C2PA/EU AI Act labelling, in-browser AI models, AI search traffic decline, AI-detector bias, LLM JSON failures, hidden prompt injection, AI image metadata, and AdSense/AI content policy. Results were read critically; vendor-authored and aggregator pages were discounted.

Rating	Meaning
High	Peer-reviewed or official primary source, or several independent sources agree.
Medium	Consistent across multiple secondary sources, but no primary source reviewed, or sources are partly commercial.
Low	Single source, vendor-authored, or contradictory figures. Use as a lead to investigate, not as a decision input.

3.2 Evidence register

Dated 11 November 2026. Figures are quoted as the sources reported them and have not been independently reproduced.

ID	Finding	Confidence	Implication for the product
E1	Incumbent PDF tools (iLovePDF, Smallpdf) generally process files on their servers. Reported retention windows differ by source (roughly 1–2 hours). Most “privacy risk” articles found are written by sellers of browser-only rivals.	**Medium** (architecture) / **Low** (retention details)	Local processing is a legitimate positioning angle, but we must prove it and avoid exaggerating rivals’ risks.
E2	Surveys of how many workers paste sensitive data into AI range from about 18% to 79%, depending on definition. Examples: Zapier (May 2026) 79% entered sensitive info; TELUS (Jan 2025) 57% entered confidential data; Kolmogorov Law (Jul 2026) 38% used personal AI accounts for work info; LayerX reported ~40% of uploaded files contained PII or payment-card data. The widely repeated “77%” is often misreported.	**Medium** (direction) / **Low** (magnitude)	The behaviour is real and common. A local redactor addresses a real risk; do not quote any single percentage in marketing.
E3	PDF→Markdown for LLM/RAG is served by hosted APIs and Python libraries (PyMuPDF4LLM, Marker, Docling, MinerU). Tables are the fragile part: Markdown cannot express merged cells, so MinerU emits HTML tables. No clearly browser-only tool with local OCR surfaced in the search.	**Medium** (tooling) / **Low** (“no browser tool exists” — absence of evidence)	Gap candidate for a no-install, no-upload converter with an honest quality report. Must verify no incumbent already does this well.
E4	Many token counters exist. Counts are exact only for OpenAI tokenizers; other providers are estimates (one vendor claims 5–15% error). One source claims newer Claude/Gemini tokenizers produce ~30% more tokens than expected on some text.	**Medium**	Crowded. The only defensible edge is honesty (Exact/Estimated labels, dated price data) and extras (context-fit, chunking) — not another counter.
E5	EU AI Act Article 50 transparency duties reportedly apply from 2 Aug 2026, with a later transitional date for existing systems. The Act is technology-neutral; C2PA is the named example in a draft Code of Practice. C2PA metadata is easily stripped on re-export.	**Medium** — primary text (Regulation (EU) 2024/1689) **not reviewed**	There is demand for inspecting provenance. We must not give legal advice and must have legal review before publishing any compliance wording.
E6	AI images carry readable metadata: Stable Diffusion PNGs store prompts and settings in text chunks; C2PA manifests live in JPEG APP11 / PNG caBX / WebP chunks; IPTC “trainedAlgorithmicMedia” drives “Made with AI” labels. Many strippers exist; some are explicitly built to evade AI-disclosure labels.	**Medium**	A privacy-focused inspector/cleaner is useful. Tools that exist to defeat disclosure or watermarks are an ethical and legal line we will not cross (§3.4).
E7	In-browser AI is practical: Transformers.js v4 (Feb 2026) with WebGPU runs background removal (MODNet, RMBG) and Whisper tiny/base. OCR evidence is thin — one hobby project only.	**Medium** (bg removal, Whisper) / **Low** (OCR)	On-device AI tools are feasible but need model-licence, size and device gates. OCR needs its own prototype before commitment.
E8	LLM “JSON mode” still fails: truncated output, valid JSON that violates the schema, wrong types. Common fixes are jsonrepair-style syntax repair, schema validation and bounded retries. A repair step can silently invent values to satisfy a schema.	**Medium** (blogs and docs; consistent)	A repair tool that **shows every change** and never auto-fills missing required fields is a clear, defensible improvement.
E9	A USENIX Security 2026 measurement found hidden prompt injections in about 1% of ~197,000 resumes on a production hiring platform. Detection works on PDF structure (near-white text, text under ~2 pt) but is bypassable (render mode 3, hidden layers, embedded JS).	**Medium** (one study via secondary coverage)	Real, narrow need among recruiters, educators and RAG builders. The tool must never say “safe” — only “no known patterns found”.

ID	Finding	Confidence	Implication for the product
E10	AI-text detectors are biased against non-native English writers: a Stanford study (2023, TOEFL essays) found over 61% misclassified by seven detectors; Waterloo discontinued Turnitin's AI detection in Sept 2025.	**Medium–High**	Do **not** build an AI-text detector. It would harm real users and its accuracy cannot be defended.
E11	AI Overviews cut click-through on affected queries: Seer 1.76%→0.61%; Pew 15%→8%; Ahrefs −58% for the top result. Chartbeat: small publishers lost ~60% of search referrals. **None of this is specific to tool or calculator sites.**	**Medium** (direction) / **Unknown** (effect on tool queries)	Plan for volatile traffic. Hypothesis H1: "do" queries (compress, merge, fix) are more resilient than "what is" queries. Measure it.
E12	Google's "scaled content abuse" spam policy (March 2024) targets mass-produced low-value pages; AdSense separately prohibits ads on low-value content. No source addressed tool sites specifically; Google's own pages were not reviewed.	**Low–Medium**	Reinforces v1's rule: one page per real intent, no near-duplicate format pages. Keep the AI Prep set small and each page genuinely functional.

3.3 Problem → gap → solution map

Each row is a problem a real user has today, why existing options fall short (as far as the evidence shows), and what we build. "Wave" refers to the release plan in §6.3.

ID	Problem	Gap in what exists today	Our solution	Wave
G1	People paste emails, contracts, logs and customer data into chatbots and leak personal or confidential information (E2).	Enterprise DLP exists but is heavy and company-managed. Individuals and small teams lack a simple local tool that finds sensitive items before paste.	**Safe-to-Paste Redactor** — detect PII and secrets locally, swap for placeholders, show every finding, and restore originals in the AI's reply.	1
G2	Model output is broken JSON or violates the expected schema, breaking pipelines (E8).	Generic JSON formatters stop at "invalid". Repair libraries are code, not a visual tool, and can hide invented values.	**LLM JSON Repair & Schema Check** — repair syntax, list each change, validate against a schema, and never fabricate missing fields.	1
G3	Users cannot tell if text fits a model's context window, what it will cost, or how counts differ by model (E4).	Counters are plentiful but rarely say which counts are exact versus guessed, or show data freshness.	**Token & Context Calculator** — Exact/Estimated labels, dated model and price data, context-fit bar, chunk splitter.	1
G4	Documents sent to AI contain invisible text, zero-width or Unicode-tag characters, or hidden instructions (E9).	Detection exists as research code and a few small tools; nothing mainstream for non-developers.	**Hidden Text & Invisible Character Scanner** for text, PDF, DOCX and HTML with a located, explained report.	1
G5	PDFs and Word files turn into poor input for LLMs: lost structure, mangled tables, repeated headers (E3).	Strong tools are Python libraries or hosted APIs that receive the document.	**Document → LLM-ready Markdown** in the browser, with a quality report, honest table handling, token estimate and chunking.	2
G6	Photos and AI images leak GPS, device data or generation prompts; people can't see what a file carries (E6).	Many strippers; few show everything first, re-scan the result, and handle AI/C2PA data transparently.	**Image Metadata Inspector & Privacy Cleaner** — inspect first, clean with presets, verify after.	2
G7	Upload-based tools expose sensitive files to third parties and gate features behind accounts (E1).	Incumbents' core architecture is server-side. Local tools exist but trust is unproven.	Keep v1's local-first design but add **verified, dated local-processing badges** and a "check it yourself" guide.	0 (core)

ID	Problem	Gap in what exists today	Our solution	Wave
G8	Background removal, transcription and OCR normally mean uploading private photos or audio (E7).	Browser-local versions exist as demos; none established as a trusted, supported hub tool.	**On-device AI tools** (background remover, transcriber, OCR) — only after licence, size and device gates pass.	3 (gated)
G9	Search clicks are shrinking under AI Overviews; a tool site cannot depend on informational queries (E11).	Most tool sites lean on SEO articles that AI summaries replace.	Tool-first pages for “do” queries; measure resilience (H1); build direct and return usage; keep guides for trust and quality, not as the traffic plan.	Ongoing
G10	AI text detectors falsely accuse non-native writers (E10).	Detectors are widely sold despite documented bias.	**Deliberately not built.** Public rationale published in the About/FAQ if asked.	—

3.4 What we will deliberately not build

- **An AI-text detector** (E10). Unreliable and harmful to non-native writers.
- **Watermark removers or tools marketed to defeat AI-disclosure labels** (E6, E5). Provenance stripping exists only as an explicit, opt-in privacy option with a plain-language notice (§7.6) and is never described as “remove AI label”.
- **Server-side AI inference or any hidden upload of user content.** Every AI Prep tool processes locally or does not ship.
- **Generative features** (rewriting, summarising, image generation). They would require servers or large models and blur the trust message.
- **Official-looking presets for exams or government forms** without a dated, sourced specification (unchanged from v1).

3.5 Hypotheses to validate before scaling

ID	Hypothesis	How to test (cheap first)
H1	Task (“do”) queries lose fewer clicks to AI Overviews than informational queries.	Search Console CTR by page type after 8–12 weeks; compare against a few SERPs checked monthly for AI Overview presence.
H2	A meaningful share of users choose a tool because it runs locally.	Landing-page message test; on-page “why local?” engagement; short exit survey (no content collected).
H3	Individuals (not just companies) want a local PII redactor.	Keyword and SERP check for redact/anonymize queries; prototype click-through from the Word Counter and Token tools.
H4	Searchers want “fix broken JSON from ChatGPT” specifically, not just “JSON formatter”.	Keyword research on both phrasings; check whether the SERP is tool-led or forum-led.
H5	Cross-tool use is real: users move from commodity tools into AI Prep tools.	Related-tool click events (§13.1) and returning-session rate.
H6	Photo/signature resizing is a viable niche (carried over from v1).	Keyword data for India and US, SERP review, interviews.

4. Goals, non-goals and success measures

4.1 Goals

- Launch a coherent first release: **12 core tools in six categories**, plus **4 Wave-1 AI Prep tools** in a seventh category if Decision D1 is approved.
- Complete core file, text and developer tasks without mandatory sign-up.

- Keep every workflow client-side, and **verify** the claim per tool with a network inspection before the page says “processed on your device”.
- Create crawlable, indexable pages for real search intents — one page per real intent.
- Build one maintainable framework so new tools share validated patterns, including the shared text workspace and detector engine.
- Prepare for an AdSense application only after quality, policy and content checks — and keep a fallback plan.

4.2 Non-goals for the MVP

- User accounts, cloud storage, file history or cross-device sync.
- Paid plans, subscriptions, usage credits or API access (revisit under Decision D6).
- **Generative AI** of any kind (rewriting, summarising, image generation) and any **server-side AI inference**. (Revised: v1 also excluded OCR and document understanding; those are now permitted only as gated on-device tools, §7.8.)
- AI-text detection, watermark removal, or tools designed to defeat AI-disclosure labels (§3.4).
- Full office-document conversion with layout-fidelity guarantees; advanced PDF editing, signing, redaction of PDF content, password removal.
- Dynamic QR codes, scan analytics or account-based tracking.
- Multi-language launch (build localization-ready text structure; launch English-first).

4.3 Success measures

Metric	Definition	Use
Successful task completions	Successful outputs divided by tool-start events.	North-star product health.
Organic clicks / impressions / CTR	Search Console, by page and query; **segmented by page type** (core vs AI Prep vs guide).	Acquisition, and test of H1.
Tool error rate	Failed processing or validation events divided by starts.	Reliability.
Return usage and cross-tool use	Returning sessions; related-tool clicks from core tools into AI Prep and back (privacy-conscious analytics).	Test H5; justify more tools.
Verified-local coverage (new)	Share of published tools whose “local” claim has a dated network audit on file.	Trust integrity. Target: 100% before any tool shows a local badge.
Detector quality on corpus (new)	Precision and recall of the PII, secret and hidden-text detectors on the published synthetic test corpus.	Honest accuracy statements; regression guard.
Core Web Vitals	Field LCP, INP, CLS where enough data exists.	Experience quality.
Ad revenue / page RPM	Per thousand pageviews, once approved.	Monetization — not an early proxy.

Do not set traffic or revenue forecasts until country-level query demand, ranking potential and actual monetization data exist.

5. Audience and use cases

Segment	Jobs to be done	Relevant tools
Students and applicants	Resize photos for forms, combine scans, count essay words.	Photo/signature resizer, JPG to PDF, Merge PDF, Word Counter.
Office workers	Prepare documents and images; combine or extract pages.	Image Compressor, Merge/Split PDF.

Segment	Jobs to be done	Relevant tools
Developers and QA	Inspect and format payloads.	JSON Formatter.
Creators and small businesses	Lightweight images; shareable QR codes.	Compressor, Converter, Resizer, QR.
General users	Age and date maths; Wi-Fi or URL QR codes.	Age, Date Difference, QR.
AI-assisted knowledge workers (new)	Use chatbots on real work documents without leaking names, numbers or secrets; fit long documents into a model.	Safe-to-Paste Redactor, Token & Context Calculator, Doc→Markdown.
Developers building with LLMs (new)	Fix broken model output; validate against a schema; estimate cost and context.	LLM JSON Repair, Token Calculator, Doc→Markdown, Hidden Text Scanner.
Recruiters, educators, reviewers (new)	Check submitted PDFs and documents for invisible text before running them through AI tools.	Hidden Text Scanner.
People who publish photos or AI images (new)	See and remove location, device and prompt data before sharing.	Image Metadata Inspector & Cleaner.

Core user journeys

Journeys for the six v1 categories are unchanged (image, PDF, text/JSON, QR, calculator). New journeys:

- **Redactor:** paste text → see highlighted findings by type → accept or reject each → copy the safe version → use the AI → paste its reply back → restore the originals.
- **JSON repair:** paste broken output → see repaired result and a change list → optionally add a schema → see validation errors by path → copy JSON or copy a “fix this” prompt for the model.
- **Token calculator:** paste text or drop a file → see counts per model with Exact/Estimated labels → see whether it fits each context window → split into chunks.
- **Hidden text scanner:** drop PDF/DOCX/HTML or paste text → see a located report → export a cleaned text copy → remember there is no “safe” verdict.
- **Doc → Markdown:** drop a document → see a quality report and preview → copy or download Markdown → see token estimate and chunks.
- **Metadata cleaner:** drop an image → see everything it carries → pick a preset → download → see the re-scan proving what was removed.

6. Product scope, scoring and release waves

6.1 Tool inventory

Category	Tool	Priority	Core result
Image	Image Compressor	P0	Smaller raster image with preview and measured size change.
Image	Image Converter	P0	Convert among JPEG, PNG, WebP.
Image	Image Resizer	P0	Resize by pixels or percentage.
Image	Photo & Signature Resizer	P0	Meet user-entered pixel and file-size limits where possible.
PDF	Merge PDF	P0	Combine PDFs in user order.
PDF	Split PDF	P0	Extract pages or ranges.
PDF	JPG to PDF	P0	Create a PDF from images.
Text	Word & Character Counter	P0	Live text statistics.
Developer	JSON Formatter & Validator	P0	Parse, validate, format, minify.
QR	QR Code Generator	P0	Static QR codes.

Category	Tool	Priority	Core result
Date	Age Calculator	P0	Age at a chosen date.
Date	Date Difference Calculator	P0	Calendar span between dates.
AI Prep	**LLM JSON Repair & Schema Check**	**P0†**	Repair model JSON, validate against a schema, show every change.
AI Prep	**Safe-to-Paste Redactor**	**P0†**	Find and mask PII/secrets locally; restore after the AI replies.
AI Prep	**Token & Context Calculator**	**P0†**	Token counts with Exact/Estimated labels; context fit; chunking.
AI Prep	**Hidden Text & Invisible Character Scanner**	**P0†**	Located report of hidden or invisible content.
AI Prep	Document → LLM-ready Markdown	P1	Structured Markdown with quality report.
AI Prep	Image Metadata Inspector & Cleaner	P1	Show and remove metadata; verify result.
AI Prep (gated)	Background Remover, Audio Transcriber, OCR	P2	On-device models only, after gates in §7.8.
AI Prep (candidate)	AI Crawler Policy Builder	P2	Generate robots.txt rules for AI crawlers; needs a verified bot list.

† P0 only if Decision D1 is approved; otherwise P1. P0 = launch blocker, P1 = after the core is reliable, P2 = later, per v1 definitions.

6.2 Prioritization of new candidates

Scored with v1's own model: 1–5 on each factor, weighted 20% intent fit, 20% demand evidence, 20% SERP opportunity, 20% differentiation, 10% feasibility, 10% content depth. **These are my judgement scores, not measured data.** Demand and SERP scores are deliberately low-confidence because no keyword or SERP data has been collected yet; rescore after Stage 0 (§14.1).

Candidate	Intent	Demand	SERP	Diff.	Build	Depth	Score
LLM JSON Repair & Schema Check	5	3	3	4	5	4	**3.9**
Safe-to-Paste Redactor	5	3	3	4	4	4	**3.8**
Document → LLM Markdown	5	3	3	4	3	4	**3.7**
Image Metadata Inspector & Cleaner	5	3	3	3	4	4	**3.6**
Hidden Text Scanner	4	2	4	4	3	4	**3.5**
Token & Context Calculator	5	4	2	2	5	3	**3.4**
Audio Transcriber (on-device)	5	4	2	3	2	3	**3.3**
AI Crawler Policy Builder	4	2	3	3	5	3	**3.2**
Background Remover (on-device)	5	4	1	2	3	3	**3.0**

Reading the table: the Token Calculator scores lower on differentiation (crowded), yet is still in Wave 1 because it is cheap, shares the text workspace and feeds the other tools. Background removal is high-demand but the SERP is crowded and it needs a large model — hence Wave 3.

6.3 Release waves

Wave	Scope	Gate to proceed
0 Core	The 12 v1 tools, unchanged in intent.	Test corpus passes; local-processing claims audited.

Wave	Scope	Gate to proceed
1 AI Prep text tools	LLM JSON Repair, Safe-to-Paste Redactor, Token & Context Calculator, Hidden Text Scanner (text and DOCX/HTML first; PDF structure checks if the library prototype succeeds).	Shared text workspace works on a low-power mobile; detector corpus precision/recall published; no network calls verified.
2 Document tools	Doc → LLM Markdown (text-layer PDFs, DOCX); Image Metadata Inspector & Cleaner.	Wave-1 pages show impressions or engagement; table-handling quality accepted on the corpus.
3 On-device AI (gated)	OCR, Background Remover, Audio Transcriber; AI Crawler Policy Builder.	All gates in §7.8 passed per tool; evidence from Waves 1–2 that AI Prep earns traffic.

6.4 Global requirements for every tool

All v1 global requirements remain (dedicated URL and copy, validation before work, no duplicate jobs, predictable filenames, reset and resource release, no content in analytics, keyboard access, visible limits from configuration). Added in v2:

- **Accuracy label.** Any computed result that could be wrong shows **Exact**, **Estimated** or **Heuristic**, with a one-line explanation.
- **Show what changed.** Any automatic edit (repair, redaction, cleaning, conversion) is shown as a list or diff the user can review before copying or downloading.
- **No “safe” verdicts.** Detection tools report what they found and what they cannot detect. Wording such as “secure”, “anonymised” or “safe” is banned unless provable.
- **Local-processing proof.** The “runs on your device” badge appears only when the registry holds a dated network-audit record (§9.3). The page links to a short guide that shows users how to check it themselves.
- **Dated data.** Pricing, model lists and bot lists show their review date and a staleness banner after the configured maximum age.
- **Cross-linking.** Core tools link to the AI Prep tool a user is likely to need next (for example Word Counter → Token Calculator; JSON Formatter → LLM JSON Repair; image tools → Metadata Cleaner). Links must be relevant, not a link dump.

7. Tool-by-tool specifications

7.1 Core 12 tools (condensed from v1)

The full v1 requirement set is retained. This table keeps the decisions and acceptance essentials in one place; v1 §5 remains the detailed reference where a team needs more. Changes are marked **NEW**.

Tool	Key requirements	Acceptance essentials
Image Compressor	JPEG/PNG/WebP in; quality slider for lossy outputs only; optional resize; batch with ZIP; before/after preview with size, % saved, dimensions. Warn if output is larger. Web Worker. **NEW:** option “remove metadata (recommended)” on by default; link to Metadata Cleaner.	Corpus: transparent PNG, photo JPEG, WebP, large, portrait/landscape, batch. Output decodes and matches chosen dimensions/format. Never show a quality slider that does nothing for PNG.
Image Converter	Among JPEG/PNG/WebP; background colour when converting to JPEG; quality only for lossy targets; batch with per-file status.	Transparency, EXIF orientation, very wide/tall, MIME/extension consistency.
Image Resizer	Width, height, percent, aspect lock, contain fit; reject zero/negative/huge; show resulting size.	Aspect ratio, up/down scaling, rounding, transparency.
Photo & Signature Resizer	Custom width/height and max KB/MB; iterative compress/resize; explain if target cannot be met and show the closest result. **No hard-coded exam/government presets** without source, date, owner.	Exact pixel dimensions and byte size verified. Never label “compliant” unless it matches the user’s entered rules. Niche demand still unverified (H6).

Tool	Key requirements	Acceptance essentials
Merge PDF	Reorder (drag + keyboard), remove, page counts; client-side library; limits shown first; warn on encrypted/malformed files. **NEW:** option to drop document metadata.	Mixed page sizes, rotations, protected/malformed files, large files, order.
Split PDF	All pages, specific pages or ranges (1-3, 5, 8-10); validate range syntax; summary of outputs; ZIP when multiple.	First/last page, non-contiguous, blank, rotated, invalid syntax.
JPG to PDF	Reorder; A4/Letter/fit; orientation; margins; fit mode. Title may be “Images to PDF” if PNG supported.	Mixed dimensions, EXIF orientation, long images, rendering in independent viewers.
Word & Character Counter	Words, characters with/without spaces, sentences, paragraphs, reading and speaking time; “How we count” explainer; copy, .txt download, remove extra spaces. **NEW:** “Estimate tokens” link to Token Calculator.	Empty text, whitespace types, punctuation, CJK, apostrophes, emoji, line breaks.
JSON Formatter & Validator	Format, validate, minify, copy/download; error with line/column when available; sample JSON; no transmission. **NEW:** when parsing fails and the text looks model-generated, offer “Try LLM JSON Repair”.	Valid/invalid, escapes, trailing commas, nesting, Unicode, large input.
QR Code Generator	URL, text, Wi-Fi, email, phone, vCard; colours with contrast warning; PNG and SVG; “test-scan before printing”. No payload to analytics.	Multiple sizes, contrast levels, real phones, long URLs, Unicode, Wi-Fi special characters.
Age Calculator	Calendar-date arithmetic (never ms ÷ 24 h); define Feb-29 and month-end rules; reject future birth dates.	Leap years, Feb 29, month ends, identical dates, timezone boundaries.
Date Difference	Date-only model; exclusive by default with explicit inclusive option; explain reversed dates.	Same day, reversed, leap year, month end, year boundary, DST.

7.2 LLM JSON Repair & Schema Check (/tools/llm-json-repair)

Problem (G2, E8): model output is often truncated, wrapped in code fences, or valid JSON that does not match what the program expects. **User outcome:** a valid, trustworthy JSON object — and certainty about what was changed.

Area	Requirement
Inputs	Pasted text or .json/.txt file. Optional JSON Schema (paste or file); initial support for the common draft 2020-12 keywords (type, properties, required, enum, items, minimum/maximum, pattern, oneOf/anyOf, additionalProperties). Unsupported keywords are listed, not ignored silently.
Syntax repair	Remove markdown code fences and surrounding prose; fix trailing commas, single quotes, unquoted keys, comments, Python/JS literals (None, True, undefined, NaN), unescaped newlines in strings; close truncated objects/arrays/strings. Deterministic and reproducible.
Change list	Every repair is listed with position, rule name and before/after. User can undo individual repairs. Result is labelled **Repaired** (syntax fixed) or **Unchanged**.
Truncation	If output looks cut off, show a prominent notice: “Data after the cut is missing. Repair only closes brackets.” Never invent values for incomplete entries.
Schema check	After syntax is valid, validate against the schema. Report each error with JSON path, expected, actual. Separate **Syntax repaired** from **Schema violation — needs regeneration**. **Never auto-fill missing required fields.**
Extras	“Copy fix prompt”: generates plain text listing the errors for the user to send back to the model. Format, minify, copy and download. Link to JSON Formatter.
Processing	Entirely in the browser; no eval/Function on user input; parser worker for large input.
Accuracy label	Syntax rules: **Exact** (deterministic). Schema validation: **Exact** for supported keywords. Truncation detection: **Heuristic**.
Acceptance	Corpus of at least 50 realistic malformed outputs (fences, truncation, comments, mixed quotes, Python literals, nested errors) with expected results. Repairs are idempotent. Repaired output always parses with a strict parser. Large-input timing recorded on a lower-powered phone.

Area	Requirement
Risks	Repair that changes meaning (e.g. choosing between ambiguous quotes) → show warning on ambiguous rules and default to the least-invasive option.

7.3 Safe-to-Paste Redactor (/tools/pii-redactor)

Problem (G1, E2): people paste real data into chatbots. **User outcome:** a version of the text with sensitive items masked, a clear list of what was found, and the ability to restore the originals in the AI's reply.

Area	Requirement
Inputs	Pasted text (P0); .txt/.md/.csv/.log files (P0). DOCX and text-layer PDF are P1.
Detectors (P0, deterministic)	Email; phone numbers (E.164 and common national formats); payment-card numbers (Luhn-validated); IBAN (checksum-validated); IPv4/IPv6; URLs containing tokens or credentials; API keys and tokens by well-known prefixes and structure (e.g. cloud-provider access keys, JWTs, private-key headers); national ID formats **only where a checksum exists**, listing the supported countries.
Detectors (P1, model-assisted)	Names, organisations and addresses need an on-device NER model (§7.8 gates apply). Until then the page states plainly that **names and free-form addresses are not detected**.
Masking	Replace with typed placeholders such as [EMAIL_1], consistent per value. User can accept, reject or add manual selections per finding. Choose placeholder style (labelled, or generic).
Restore	Paste the AI's reply; placeholders are swapped back using the mapping held **in memory only**. Mapping is never written to storage, URL, logs or analytics. Optional "download mapping" is explicit and warns that the file contains the sensitive values.
Report	Counts by type; list of unchecked categories; precision/recall on the public corpus linked from the page.
Wording rules	Never use "anonymised", "secure" or "compliant". Required notice: "This tool finds common patterns. It can miss things. Review the result before sharing."
Accuracy label	Pattern + checksum detectors: **Exact for the pattern**. Everything else: **Heuristic**.
Acceptance	Synthetic corpus with seeded PII, near-miss decoys (order numbers, long digit strings) and non-English samples. Published precision and recall per detector. Zero network requests during use (verified). Large pasted text stays responsive. Mapping cleared on reset and tab close.
Risks	False sense of security → prominent limitation copy and "not detected" list. False positives on legitimate numbers → per-finding toggle.

7.4 Token & Context Calculator (/tools/token-counter)

Problem (G3, E4): knowing whether text fits a model and what it costs, without being misled by false precision. **User outcome:** trustworthy counts and a plain answer to "will this fit?".

Area	Requirement
Inputs	Pasted text or .txt/.md/.json/.csv files. P1: text extracted from DOCX/PDF via the Doc → Markdown pipeline.
Model data	A versioned data file lists, for each model: name, context window, input/output price per million tokens, tokenizer family, and **review date**. Stored in the repository, not fetched from a server at runtime. Banner when older than the configured maximum age (default 30 days). Each row links to the vendor's pricing page.
Counting	Use locally runnable open tokenizers where licences allow. Where the vendor's exact tokenizer is not available offline, show a range and the label **Estimated** with the method. **The tool never sends text to a vendor API to "get an exact count".**
Outputs	Token count per model; characters, words, lines; cost for N calls; a context-fit bar (fits / tight / exceeds, leaving room for a user-set output size); optional view highlighting token boundaries.
Chunk splitter	Split by token budget with optional overlap, preferring paragraph and heading boundaries; download as files or copy chunks.
Extras	Language/code comparison: show how the same text costs differently across tokenizers, with the caveat that English prose differs less than code or structured data.

Area	Requirement
Accuracy label	**Exact** only where the exact tokenizer runs locally; otherwise **Estimated** with margin and method.
Acceptance	Known-answer tests against published tokenizer outputs for each locally supported tokenizer; Unicode, emoji, CJK, long-line and code inputs; large text responsiveness; every price row has a review date and source link.
Risks	Data goes stale → owner, monthly review, staleness banner, CI check that fails the build when data is older than the limit. Crowded market → the edge is honesty and the chunker, not the counter.

7.5 Hidden Text & Invisible Character Scanner (/tools/hidden-text-scanner)

Problem (G4, E9): documents can contain text that people cannot see but AI systems read. **User outcome:** a located report of anything hidden or invisible, and a cleaned text copy.

Area	Requirement
Inputs	Pasted text (P0); .txt/.md/.html (P0); .docx (P0); text-based PDF (P0 if the Wave-1 prototype succeeds, otherwise P1). File type is verified by content, not extension.
Text checks	Zero-width characters; bidirectional control characters; Unicode “tag” characters (U+E0000 block); unusual whitespace; mixed-script lookalikes (homoglyph risk flagged, not asserted); invisible separators.
PDF checks	Near-white or background-matching text; very small text (under about 2 pt); invisible rendering mode; text outside the page; content in optional layers that are off by default; annotations; embedded JavaScript (reported, never executed).
DOCX / HTML checks	DOCX: hidden-text property, white or tiny text, comments, tracked-change remnants. HTML: display:none, hidden attributes, off-screen positioning, comments.
Instruction heuristics	Optional: flag phrases that look like instructions to an AI system. Labelled **Heuristic**, may produce false positives, off by default.
Report	Each finding: type, location (page/paragraph/offset), the hidden text, why it was flagged. Visible-text vs extracted-text comparison. Export cleaned plain text.
Wording rules	Result is “**N findings**” or “**No known patterns found**” — never “safe”. Required note: “Some techniques cannot be detected by this tool.”
Accuracy label	Character-level checks: **Exact**. Structural PDF/DOCX checks: **Exact for the rule, incomplete overall**. Instruction phrases: **Heuristic**.
Acceptance	Corpus of crafted files covering every check, each with a clean control; no execution of any embedded content; malformed and huge files fail safely with a clear message; zero network calls.
Risks	Determined evasion (known bypasses exist) → honest limitations, link to guidance that untrusted documents should be treated as untrusted input regardless of scanning.

7.6 Image Metadata Inspector & Cleaner (/tools/image-metadata-cleaner)

Problem (G6, E6): images carry location, device, and sometimes generation prompts and provenance data users cannot see. **User outcome:** know exactly what a file carries, remove what they choose, and verify the result.

Area	Requirement
Inputs	JPEG, PNG, WebP (single file P0, batch P1). Content-based type check.
Inspect	EXIF (GPS coordinates shown as text only, with no map lookup so nothing is sent anywhere; device make/model, serial numbers, timestamps, software), XMP, IPTC, PNG text chunks (including generator settings and prompts), and **presence** of a C2PA/Content Credentials manifest with its claim generator and a clear “signature not verified by this tool” label. Full C2PA signature validation is P2 and needs a trust-list decision.
Clean presets	**Privacy (default):** removes GPS, device and serial data, timestamps, and prompt/generator text. **Custom:** per-field toggles. **Remove everything including provenance:** separate opt-in with a plain-language notice — “Content credentials can show where an image came from. Some platforms and laws expect AI-generated content to stay labelled. You are responsible for how you use the result.”

Area	Requirement
Method	Prefer lossless removal (JPEG segments, PNG chunks) over re-encoding. Preserve colour profile and orientation unless the user removes them. Warn when re-encoding is unavoidable.
Verify	After cleaning, re-scan the output and show a before/after table. Download with predictable filename.
Hard limits	No invisible-watermark removal. No feature, copy or SEO text that positions the tool as defeating AI labels. Review marketing text and keyword targets against this rule.
Accuracy label	Field presence: **Exact**. “Removed everything”: **never claimed** — only “removed the fields listed”.
Acceptance	Corpus: camera JPEG with GPS, phone HEIC-origin JPEG, Stable-Diffusion-style PNG with text chunks, C2PA-signed sample, WebP with XMP. Output decodes, matches expected pixels, and the re-scan shows the removed fields gone.
Risks	Dual use (evading disclosure) → default privacy preset, opt-in friction, legal review of notice text before publishing (E5).

7.7 Document → LLM-ready Markdown (/tools/pdf-to-markdown)

Problem (G5, E3): documents lose structure when converted for AI use. **User outcome:** clean, structured Markdown plus an honest account of what may be wrong.

Area	Requirement
Inputs	Text-layer PDF, DOCX, HTML, TXT (P0 within this tool). Scanned PDFs require on-device OCR and wait for the §7.8 gates (P2). Password-protected files: clear message, no workaround.
Extraction	Headings inferred from structure/font size; lists; paragraphs re-joined across line breaks; repeated headers/footers and page numbers removed (toggle); page markers optional; links kept; images listed as omitted.
Tables	Simple tables → Markdown pipe tables. Tables with merged cells or uncertain structure → HTML table or CSV with a visible “structure may be lossy” note. Never silently flatten.
Quality report	Pages with no text layer; suspected multi-column pages; tables detected and how each was rendered; characters that could not be mapped; reading-order confidence.
Outputs	Markdown (copy/download); optional JSON with page references; token estimate (via §7.4 data); chunk splitter by headings and token budget.
Safety	Runs the §7.5 hidden-text checks and shows a one-line result. Never executes embedded scripts.
Accuracy label	Structure inference: **Heuristic**. Text content: **Exact** for text-layer PDFs; reading order: **Heuristic**.
Acceptance	Corpus: single-column report, two-column paper, table-heavy financial statement, rotated pages, ligature-heavy text, large file. Manual review checklist plus automated text-recall comparison to a reference extraction. Limits and memory tested on a lower-powered phone.
Risks	Hard layouts will fail → say so up front; do not promise fidelity. Prototype library choice for size, licence and worker support before committing.

7.8 Gated on-device AI tools (Wave 3, P2)

Background Remover, Audio Transcriber (Whisper-class), and OCR (also feeds §7.7 and the Redactor’s name detection) are feasible in the browser (E7) but are not committed until each passes every gate below.

Gate	Requirement
Model licence	Commercial use explicitly allowed for the exact model weights; licence text, source and review date stored in the registry. Some popular background-removal weights are reported to carry non-commercial terms — **check each model before use**.
Size and consent	Download size shown before fetching; user clicks to download; cached with a visible “remove downloaded model” control; works on a metered-connection warning.
Hosting cost	Bandwidth cost per thousand first uses estimated; model files served from our own origin or a licensed CDN (so no third party sees usage).

Gate	Requirement
Device support	WebGPU path plus WASM fallback; minimum supported devices documented; graceful message where unsupported; memory and temperature tested on a lower-powered phone.
Quality	Published test set and pass criteria; clear statement of known failure modes (hair, glass, accents, noise).
Privacy	Network audit proves no content leaves the device after the model has loaded; CSP updated for WASM only as needed.
Evidence	Waves 1–2 show that AI Prep pages earn impressions or engagement.

AI Crawler Policy Builder (P2 candidate): generates robots.txt rules for AI crawlers. Requires a maintained, sourced list of crawler names with review dates; otherwise it will mislead. No model needed.

8. Experience, visual system, sitemap and URLs

8.1 Visual direction (unchanged)

Playful, not childish: friendly utility aesthetic with rounded cards, reassuring microcopy and clear hierarchy. No novelty animation that slows a task.

Token	Direction
Primary	Deep teal (#087E8B) for trust and primary actions.
Secondary / accent	Warm coral (#F26B5B) for emphasis and selected states; soft yellow (#F4C95D) for tips.
Neutrals	Off-white canvas, white surfaces, dark ink (#172033), muted grey (#596579).
Type and shape	Highly legible sans-serif; consistent rounded radius scale. Motion limited to small state changes; honour reduced-motion.
New: status language	Three consistent badges used across all tools — **Exact**, **Estimated**, **Heuristic** — plus a **Verified local** badge. Colour is never the only signal (icon + text). Badges are shown with the result, not hidden in a tooltip.

8.2 Layout and tool workspace

Page types and the workspace anatomy (intro, input zone, controls, processing, result, trust and learning) are unchanged from v1. Responsive rules (single column on mobile, side-by-side on desktop only when it helps, 200% zoom, keyboard, reduced motion) are unchanged. v2 adds one block to the workspace: **“What changed”** — a review panel between processing and result for tools that edit content (repair, redact, clean).

8.3 Sitemap additions

URL	Purpose
/categories/ai-prep-tools	AI Prep category: who it is for, how to choose, honest limits
/tools/llm-json-repair	LLM JSON Repair & Schema Check
/tools/pii-redactor	Safe-to-Paste Redactor
/tools/token-counter	Token & Context Calculator
/tools/hidden-text-scanner	Hidden Text & Invisible Character Scanner
/tools/pdf-to-markdown	Document → LLM-ready Markdown (Wave 2)
/tools/image-metadata-cleaner	Image Metadata Inspector & Cleaner (Wave 2)
/guides/verify-local-processing	How to check, in your browser, that a tool does not upload your files
/accuracy	What Exact / Estimated / Heuristic mean on this site; detector test corpora and results

All v1 URLs remain. **URL and indexing rules are unchanged:** lowercase hyphenated stable slugs; self-referencing canonicals; server-rendered key content with a client-hydrated tool; no near-duplicate pages per format or per model; sitemap from published, canonical, indexable pages only; structured data only where it accurately describes visible content.

AI Prep page-count discipline (E12): do not generate one page per model ("GPT token counter", "Claude token counter"…) at launch. One Token Calculator page with a model selector, until query data shows distinct intents with distinct interface needs.

9. Component and application architecture

9.1 Folder structure (additions to v1)

```
src/
  features/
    ai-prep/
      json-repair/       # repair rules, change log, schema check
      redactor/          # detector registry, masking, restore map (memory only)
      tokens/            # tokenizer adapters, model data loader, chunker
      hidden-text/       # text, pdf, docx, html scanners
      doc-to-markdown/   # extraction, table handling, quality report
      image-metadata/    # parsers, cleaners, re-scan
  lib/
    text-workspace/      # shared paste/file input, large-text handling
    detectors/           # shared detector engine + test-corpus runner
    models/              # gated on-device model loader (consent, cache, fallback)
    data/                # dated data files: models, prices (with review dates)
  workers/
    text.worker.ts  detectors.worker.ts  model.worker.ts
```

9.2 Shared components added

Component	Responsibility
TextWorkspace	Shared paste/file input with size limits, large-text virtualisation, clear/reset, memory release. Used by Word Counter, JSON tools, Redactor, Token Calculator, Hidden Text Scanner.
FindingsList / ChangeLog	Accessible list of findings or changes with accept/reject, jump-to-location and keyboard operation. The "What changed" panel.
AccuracyBadge / LocalBadge	Renders Exact/Estimated/Heuristic and Verified-local from registry data. Cannot be rendered with hand-typed text.
DataFreshnessBanner	Shows review date of dated data and the staleness warning.
ModelConsentDialog	Size, licence, purpose; download, cache and remove controls (Wave 3).
DetectorEngine	Registry of named detectors with id, version, pattern, validator, test cases; one runner used by both the UI and the corpus tests.

9.3 Tool registry v2

v1's typed registry stays the single source of truth. v2 adds fields so that privacy and accuracy claims are enforced by code rather than by habit: the Verified-local badge renders only when an audit record exists, and the build fails if a tool with dated data has expired data.

```
type ToolDefinition = {
  id: string; slug: string; category: ToolCategory; name: string;
  shortDescription: string; title: string; metaDescription: string;
  status: "draft" | "published";
  processing: "local" | "local-model";          // no "server" value in MVP
  audit?: { verifiedAt: string; method: "network-inspection"; by: string };
  accuracy?: "exact" | "estimated" | "heuristic";
  dataAsset?: { name: string; reviewedAt: string; maxAgeDays: number };
  modelAsset?: { id: string; bytes: number; license: string; licenseReviewedAt: string };
  relatedToolIds: string[];
  acceptedMimeTypes?: readonly string[];
  maxFileBytes?: number;                        // configured and tested per tool
  component: React.ComponentType;
};
```

9.4 State and processing (v1 rules retained)

- Explicit states: idle, selected, validating, processing, success, partial-success, error, cancelled. Typed adapters return outputs, metadata, and typed errors.
- Web Workers for heavy work; object URLs and large buffers released on reset and unmount; content never stored in localStorage, cookies or analytics.
- File-signature checks, never the extension alone; feature code isolated so one library upgrade cannot break another category.
- **New:** detectors and repair rules are pure functions with versioned ids, so the UI, the test corpus and the “accuracy” page all run the same code.
- **New:** the restore mapping in the Redactor lives in a single in-memory object that is zeroed on reset, navigation and tab close, and is excluded from error reporting.

10. Technical specifications and acceptance

10.1 Recommended stack

Layer	Recommendation	Notes
Framework	Next.js App Router + React + TypeScript	Pin a supported stable version at kickoff and test compatibility.
Styling	Tailwind CSS + shared tokens	Playful visual system; responsive primitives.
Validation	Zod or equivalent	Dimensions, dates, QR fields, detector settings.
PDF	Evaluate pdf-lib (write) and a text-extraction library (read)	Prototype large files, encryption, metadata, text-position data (needed for hidden-text and Markdown tools).
Image	Browser APIs + evaluated WASM option	Benchmark quality, formats, size, worker support, mobile memory. Add a metadata parser for EXIF/XMP/IPTC/PNG chunks.
JSON	CodeMirror 6 or lighter editor; evaluated repair approach; a JSON Schema validator	Check licences, size and maintenance. Repair must be deterministic and expose its change log.
Tokenizers	Evaluated open tokenizer packages that run in the browser	Check licence and bundle size; lazy-load per tokenizer family.
On-device AI	Evaluated browser inference runtime with WebGPU + WASM fallback	Only for Wave 3 after gates. Do not add to the base bundle.
QR	Mature library, PNG and SVG	Real-device scanning; licence review.
Content	MDX first; CMS optional	Adopt a CMS only when publishing workflow requires it.

Layer	Recommendation	Notes
Hosting	Edge-capable static hosting	Confirm no reliance on serverless upload limits; size model-file hosting for Wave 3.
Testing	Vitest + Playwright + axe, plus corpus runner	Unit, browser workflows, accessibility, detector precision/recall regression.

10.2 Processing architecture decisions

Concern	MVP decision
Uploads	No server upload for any tool. Verify via network inspection and store the audit date in the registry.
Limits	Per-tool file, page and token limits from device tests; shown before processing; configuration, not scattered constants.
Browser support	Document after testing Chrome, Firefox, Safari and relevant mobile browsers. WebGPU is optional; WASM fallback required for any model tool.
Memory	Test on low-memory phones; batch and release buffers; recoverable failure message.
Filenames and downloads	Sanitise output names; Blob URLs; ZIP only for multi-file output.
Offline	Not promised in MVP.
Server APIs	None for local tools. Any future endpoint needs a justified feature and disclosed data flow.
Dated data	Model, price and bot lists live in versioned files with review dates; CI fails when data exceeds its maximum age.

10.3 Non-functional requirements

Area	Target
Performance	LCP ≤ 2.5 s, INP ≤ 200 ms, CLS ≤ 0.1 at the 75th percentile — targets, not guarantees. Heavy libraries are lazy-loaded per tool.
Accessibility	WCAG 2.2 AA: labels, focus, contrast, keyboard, errors and announcements; findings lists fully operable without a pointer.
Reliability	No silent failure; recoverable errors and clear reset for every workflow.
Security	HTTPS; dependency and licence review; restrictive CSP (WASM only where needed); no eval of user input; nothing uploaded is executed.
Privacy	No content in logs or analytics; accurate disclosures; no overclaiming.
SEO	Indexable HTML, unique titles/descriptions, canonicals, internal links, sitemap, correct status codes.
Maintainability	Strict TypeScript, typed registry, isolated features, no any without a documented exception.
Observability	Aggregate error codes and timing only — never content or filenames.

10.4 Cross-tool acceptance checklist

- Primary workflow works by keyboard alone; every control has a visible label and focus indicator.
- Loading, success, warning, partial-success and error states are designed; cancel/reset works where supported.
- Large input measured on at least one lower-powered phone.
- Network inspection confirms the stated processing location; **audit date recorded in the registry**.
- No payload reaches analytics or third parties.
- Output is reopened, rescanned or revalidated to prove it is valid.
- **Every Exact/Estimated/Heuristic label is justified by a test or documented method.**
- **Every automatic change is visible in the “What changed” panel and reversible before export.**

- Unique metadata, useful text, working internal links; no ad interferes with selection, processing or download.

11. Privacy, security, accessibility and quality

All v1 requirements are retained: plain-language processing explanation, accurate vendor list, no content in databases/logs/localStorage/session replay (or full masking of tool areas), no “files never leave your device” claim until audited, a privacy contact; documented file types and enforced limits; content-based validation; never executing macros, scripts or embedded PDF JavaScript; current dependencies; restrictive CSP; programmatic labels, keyboard operation, live regions, non-colour indicators, reduced motion; original, reviewed content with no fabricated authors, testimonials or statistics.

11.1 Threat model for AI Prep tools

Threat	Why it matters here	Mitigation
Users trust a redactor too much	A missed name or ID gets pasted into an AI; harm is the user’s.	Limitation copy beside the result; “not detected” list; per-finding review; published precision/recall; banned words (§6.4).
Sensitive data in memory or crash reports	Restore map and pasted text are sensitive.	Memory-only map; scrub on reset; error reporting captures codes only; session-replay tools disabled or fully masking.
Malicious uploaded file	Scanners parse untrusted PDFs, DOCX and HTML.	Parse in a worker; never render HTML into the page; never run scripts; size/page/time limits; fail closed.
Zip bombs / huge files	Memory exhaustion on phones.	Pre-checks on declared sizes; streaming where possible; recoverable failure.
Stale data presented as current	Wrong prices or model limits mislead decisions.	Review dates, staleness banner, CI age check, vendor links.
Dual use of metadata/provenance removal	Could be used to evade disclosure duties.	Inspect-first design; default privacy preset; opt-in friction and notice; no watermark removal; marketing review; legal review.
Compliance wording errors	EU AI Act and similar rules are easy to mis-state.	No legal claims without counsel review; link to primary text; avoid “compliant”.
Model supply-chain risk (Wave 3)	Third-party weights and runtimes.	Pin versions and hashes; host ourselves; licence review; subresource integrity where applicable.

12. Research plan: competitors, SEO and distribution

> **STATUS** This is still a **plan with partial evidence**, not a completed audit. Section 3.2 holds what the 12 searches found. Everything below must be confirmed by hands-on testing and current primary sources before it informs launch decisions.

12.1 Competitor and reference shortlist

Category	Candidates to validate	What to inspect
Image	TinyPNG, Squoosh, iLoveIMG	Quality, formats, local vs upload, batch, controls, limits, mobile flow, monetization.
PDF	iLovePDF, Smallpdf, PDF24 Tools	Merge/split workflow, upload handling, free limits, account gates, output fidelity, ads/upsell.

Category	Candidates to validate	What to inspect
Text / JSON	WordCounter.net; JSONLint, JSONFormatter.org	Counting rules, metrics, privacy wording; error detail, tree view, speed, keyboard access.
QR / date	QR Code Monkey, QR Code Generator; timeanddate.com	Static vs dynamic, scan reliability; leap-day and inclusive/exclusive conventions.
AI Prep: redaction	Open-source PII libraries and any browser-based redactors surfaced by SERP review	Detector coverage, false positives, local vs server, restore feature, honesty of claims.
AI Prep: documents	PyMuPDF4LLM, Marker, Docling, MinerU, Microsoft MarkItDown; hosted converters	Table handling, scanned-page support, browser/no-install options, privacy.
AI Prep: tokens	OpenAI's Tokenizer and multi-model counters	Which counts are exact, data freshness, context-fit features, local vs server.
AI Prep: JSON	jsonrepair-style libraries and online repair tools	Change visibility, truncation handling, schema checks, whether they invent values.
AI Prep: metadata	C2PA/Content Credentials viewers and metadata cleaners	What they show, validation depth, local vs upload, stance on provenance removal.
AI Prep: on-device	Browser demos of background removal and Whisper	Model licences, size, device support, quality.

12.2 Hands-on test protocol (consolidated)

- Confirm the URL is active; record date, country, browser, device, viewport, researcher.
- Run one fixed corpus: photo JPEG, transparent PNG, WebP, multi-page PDF, invalid JSON, broken LLM JSON, long text, text with synthetic PII, PDF with hidden text, image with GPS, QR payload, leap-day date pair.
- Count steps from load to output; record sign-ups, paywalls, forced uploads, ads, interruptions.
- Use the network tab to learn whether content is uploaded; read the privacy notice; never trust a visual claim alone.
- Measure sizes and inspect output validity and quality; test mobile, keyboard, errors and recovery.
- Keep facts, observed behaviour, review claims and our interpretation in separate fields. Review comments only after checking dates and authenticity.

Scorecard fields and rubric are unchanged from v1 (task outcome, friction, processing, output quality, capabilities, privacy and trust, mobile/accessibility, monetization, SEO/content, evidence, opportunity; scores 1–5 for friction, quality, privacy clarity, mobile, content, differentiation). A high opportunity score is a hypothesis, not proof of traffic. Templates are in Appendix A.

12.3 SEO keyword framework

No search volume, difficulty, CPC or ranking figure has been verified. The queries below are intent hypotheses to test with current keyword tools and live SERPs, India and United States at minimum.

Cluster	Seed queries (hypotheses)	Validation note
Image compression / conversion / resizing / photo-signature	image compressor; compress JPG; reduce photo size; JPG to WebP; resize image online; signature resize; photo 20 KB	Tool-first; byte-limit modifiers; India niche unverified (H6).
PDF merge / split / JPG to PDF	merge PDF; combine PDF files; split PDF; extract PDF pages; JPG to PDF	Strong task intent; check upload-trust angle.
Text / JSON / QR / date	word counter; character count; JSON formatter; JSON validator; Wi-Fi QR code; age calculator; days between dates	Define counting rules; error explanation as a differentiator.
AI: redaction	remove personal information from text; anonymize text online; redact PII before ChatGPT; mask sensitive data for AI	Check SERP: tool-led or article-led? (H3)
AI: JSON	fix broken JSON; JSON repair; fix JSON from ChatGPT; validate JSON against schema	Compare with generic JSON formatter queries (H4).

Cluster	Seed queries (hypotheses)	Validation note
AI: tokens	token counter; ChatGPT token counter; Claude token counter; will this fit in context window; LLM cost calculator	Crowded; check for model-name intent splits before any extra pages.
AI: documents	PDF to markdown; PDF to text for LLM; convert document for RAG	Developer intent; scanned-PDF modifiers.
AI: hidden text	check PDF for hidden text; find invisible characters; zero-width character remover	Narrow; may be better as one guide plus tool.
AI: metadata	remove EXIF online; view image metadata; check C2PA content credentials	Privacy vs provenance intents differ — handle copy carefully (§7.6).

Data fields, prioritization model (weights 20/20/20/20/10/10), SERP validation workflow, page-level SEO requirements and the supporting-guide plan from v1 are retained unchanged. Guide additions: how to check a tool doesn't upload your files; what a redactor can and cannot find; understanding broken LLM JSON; what is hidden in a PDF; what your photos reveal; how token counts differ across models.

12.4 Distribution in the AI-search era

Evidence E11 shows classic organic clicks are shrinking for informational queries; its effect on tool queries is unknown. Plan accordingly:

- **Lead with tool pages** for "do" queries; treat guides as trust and quality support, not as the main traffic plan.
- **Measure H1** from the first month: Search Console CTR by page type, and a monthly manual check of target SERPs for AI Overview presence.
- **Make pages easy for AI assistants to quote correctly:** clear headings, explicit limits and formats, accurate short answers near the top. (The benefit of any AI-specific file such as llms.txt is unproven; if used, treat it as a low-cost experiment.)
- **Build direct and return traffic:** memorable name, bookmarkable pages, cross-tool links, a plain "how we handle your data" page worth linking to.
- **Avoid dependence on one channel**; see D6 and §13.4.

13. Analytics and AdSense strategy

13.1 Event taxonomy

v1 events (tool_view, tool_start, tool_success, tool_error, download, copy_result, related_tool_click) are retained with the same allowed properties. Additions:

Event	When	Allowed properties
detector_run	A detector or repair run completes	tool_id, finding_count_bucket (none/1–5/6+), detector_set_version. **Never** types of values found when they could identify a person; never content.
ai_prep_cross_use	User moves between a core tool and an AI Prep tool	source_tool_id, destination_tool_id.
model_download (Wave 3)	User consents to a model download	model_id, size_bucket, success flag.
stale_data_view	A dated-data banner is shown	data_asset_name, age_bucket. Used to catch missed reviews.

Never collect file contents, pasted text, JSON, redaction findings, QR payloads, filenames, GPS values or metadata values. Avoid properties that could identify an individual.

13.2 Ad placement principles (unchanged)

AdSense approval is not guaranteed; do a current policy review before applying and before layout changes. Ads are clearly distinguishable from controls, never interrupt upload, processing, preview or download, reserve space to avoid layout shift, use appropriate consent controls, and never run on empty, thin or broken pages. **New:** no ads inside the “What changed” panel, findings lists or the restore step of the Redactor — these are moments where a misclick has privacy consequences.

13.3 AdSense readiness gate (unchanged)

Site completeness; original useful copy; reliable tools on target browsers; current policy review from official documentation; privacy disclosures matching actual behaviour; HTTPS, no broken links, sitemap, acceptable performance; verifiable ownership; Search Console and analytics configured without sensitive inputs.

13.4 Monetization fallback (new)

Single-source revenue is a risk (E11, E12). While AdSense stays the first path, review these in parallel once usage exists — each is a proposal to evaluate, not a commitment: direct sponsorship of non-intrusive placements; affiliate links only where relevant and clearly labelled; a voluntary support option; a later paid tier for heavy batch use that keeps everything local. Any paid plan is a v1 non-goal and needs a decision (D6).

14. Delivery plan, risks and decision log

14.1 Delivery sequence and exit gates

Stage	Work	Exit gate
0. Validate	Confirm competitor shortlist; run SERP and keyword checks for the 12 core and 9 AI Prep candidates; rescore §6.2; decide domain; legal/privacy review; prototype PDF text-position extraction and JSON repair.	Evidence log complete; no unsupported forecasts; scoring updated; D1–D6 decided.
1. Foundation	Tokens, shell, directory, categories, registry v2, TextWorkspace, badges, safe analytics wrapper, sample tool.	Responsive shell and one sample tool pass design and accessibility checks.
2. Core image + PDF	Image four, PDF three (v1 stages 2–3).	Corpus passes; local claim audited; outputs open in independent viewers.
3. Core utility + AI Prep Wave 1	Word counter, JSON, QR, age, date; LLM JSON Repair, Redactor, Token Calculator, Hidden Text Scanner.	Unit and cross-browser tests; detector corpus precision/recall published; zero-network verified.
4. Content / SEO	Unique page copy, guides, /accuracy, /verify-local-processing, sitemap, canonicals.	All indexable pages valuable, with metadata and links.
5. Launch / monetization	Security, accessibility, performance audit; Search Console; analytics; policy review; AdSense application when ready.	Launch checklist signed; ads off until approved.
6. Wave 2	Doc → Markdown; Image Metadata Cleaner — if Wave-1 signals justify.	Table-handling quality and metadata verification pass.
7. Wave 3 (gated)	On-device AI tools and Crawler Policy Builder.	All §7.8 gates met per tool.

14.2 Key risks and mitigations

Risk	Impact	Mitigation
Crowded commodity market (v1)	Hard to win organic traffic.	Build on task gaps and AI Prep; validate SERPs; do not rely on breadth.
Unverified demand (v1, expanded)	Build tools nobody searches for.	Keyword and SERP evidence before Waves 2–3; Wave 1 is cheap by design.

Risk	Impact	Mitigation
AI Overviews shrink clicks (new)	Traffic and revenue fall.	Tool-first pages; measure H1; direct/return strategy; monetization fallback.
Privacy overclaim (v1)	Loss of trust or compliance exposure.	Audit-gated badge in code; no “safe/anonymised” wording; vendor inventory.
False sense of security from detectors (new)	User harm and reputational risk.	Limitation copy; published accuracy; review step; no verdict language.
Dual-use perception of metadata tool (new)	Brand and legal risk.	Inspect-first design; hard limits (§7.6); legal review; marketing review.
Stale model/price data (new)	Misleading cost and fit answers.	Review dates, CI age check, owner and cadence.
Model licensing and hosting cost (new)	Legal exposure; unexpected bandwidth bills.	Licence gate; self-hosting estimate; consent and caching.
Device fragmentation for WebGPU (new)	Features fail on many phones.	WASM fallback; capability detection; clear unsupported message.
Memory pressure; browser output differences; PDF edge cases; thin SEO pages; ad layout damage; library licensing; outdated presets (v1)	As in v1.	v1 mitigations retained.

14.3 Decision log

ID	Decision	Current choice / recommendation	Status
—	Positioning	Balanced global Everyday Tools Hub	Confirmed
—	Visual direction	Playful, friendly, trustworthy	Confirmed
—	Domain	Undecided	Revisit after naming/trademark/domain check
—	Launch language	English-first	Confirmed
—	Processing	Client-side wherever feasible	Proposed; verify tool by tool
D1	Add AI Prep as a seventh category?	**Recommended: yes**, as an experiment inside the balanced hub; Wave 1 only at launch.	**Needs your decision**
D2	Wave-1 scope	**Recommended:** 12 core + JSON Repair, Redactor, Token Calculator, Hidden Text Scanner. Alternative: 12 core only, AI Prep after launch.	**Needs your decision**
D3	Provenance-removal policy	**Recommended:** inspect-first; privacy preset default; “remove everything” opt-in with notice; never watermark removal; legal review.	**Needs your decision**
D4	On-device AI models	**Recommended:** only after all §7.8 gates; never in the base bundle.	Proposed
D5	Do we publish detector test corpora and accuracy figures?	**Recommended: yes** (/accuracy). It is the trust differentiator.	Proposed
D6	Monetization diversification	AdSense first; review fallback options (§13.4) after traffic exists.	Revisit with first traffic data
—	CMS	MDX first, CMS optional	Proposed
—	MVP tool count	12 core (+4 if D2 approved)	Revisit after validation

Appendix A. Research worksheet templates

Template	Fields
A1. Search query record	Query; country; volume source and date; SERP intent (tool/guide/mixed); top result types; AI Overview present (Y/N); opportunity evidence; decision (build/group/defer).
A2. Competitor task test	Competitor and URL; task and test-file ID; steps and friction; upload or local (verified/unclear); output quality; mobile/accessibility; evidence date.
A3. Detector corpus record	Detector id and version; test file id; expected finding; actual finding; true/false positive/negative; notes. Used to publish precision and recall.
A4. Network audit record	Tool; build id; browser; steps performed; requests observed after load; verdict; auditor; date. Required before any Verified-local badge.
A5. Dated-data review	Data asset; reviewer; sources checked; changes made; next review date.

A6. Product assumption register

Assumption	Evidence needed	Owner	Status
Users value browser-local processing (H2)	Network audit + user test	Product / Dev	Unverified
Individuals want a local PII redactor (H3)	Keyword + SERP + prototype engagement	SEO / Product	Unverified
“Fix JSON from ChatGPT” is a distinct search intent (H4)	Keyword and SERP review	SEO	Unverified
Task queries are more resilient to AI Overviews (H1)	Search Console by page type	SEO	Unverified
Users move between core and AI Prep tools (H5)	Cross-use events	Product	Unverified
Photo/signature resizing has meaningful demand (H6)	Keyword data + SERP + interviews	SEO	Unverified
AdSense is viable for the traffic mix	Country traffic + page RPM after approval	Owner	Unverified
A playful style improves approachability	Task usability tests	Design	To test

Appendix B. Launch checklist

- ☐ All launched tools are functional and pass their test corpus (12 core, plus AI Prep Wave 1 if approved).
- ☐ Every tool page has unique title, description, H1, instructions, supported formats and limits.
- ☐ No placeholder content, fake testimonials, fake statistics or unsupported performance claims.
- ☐ All links, downloads, resets, errors and mobile interactions tested.
- ☐ **Network audit record (A4) exists for every tool that shows a Verified-local badge; no badge without one.**
- ☐ **Detector corpora and results published on /accuracy; every Exact/Estimated/Heuristic label justified.**
- ☐ **Dated-data review (A5) complete; CI age check passing.**
- ☐ **Banned-wording scan passed (“safe”, “secure”, “anonymised”, “compliant”, “remove AI label”).**
- ☐ **Legal review done for provenance, EU AI Act and privacy wording.**
- ☐ Privacy policy and consent behaviour match actual vendors and practices; no tool inputs, outputs, QR payloads, findings or filenames reach analytics.
- ☐ Accessibility checks: keyboard, screen reader, focus, contrast, reduced motion.
- ☐ Performance tested on lower-powered phones; field metrics monitored after launch.
- ☐ Sitemap, robots.txt, canonicals, redirects, metadata and status codes correct; Search Console verified.

- ☐ Contact, About, Terms, Privacy and Accessibility pages complete.
- ☐ Current AdSense and consent requirements reviewed from official sources; ads off until approval and placement checks.
- ☐ Backup, rollback and error-monitoring procedures documented.

Appendix C. Sources and verification status

All sources were reached through web search on 11 November 2026. **None except the vendor's own page counts as primary evidence, and no figure was independently reproduced.** Before relying on any finding, open the primary source: Regulation (EU) 2024/1689 and Commission guidance; Google Search Central and AdSense policy pages; each competitor's current privacy policy; the Stanford and USENIX papers themselves.

Source	Link	Type	Used for
iLovePDF — GDPR/compliance page	ilovepdf.com/zh-cn/blog/pdf-compliance-gdpr-…	Vendor (primary for its own claims)	E1
Is iLovePDF Safe? (SwifDoo review)	swifdoo.com/blog/amp/is-ilovepdf-safe	Secondary; competitor-authored	E1
Best open-source PDF-to-Markdown tools 2026	themenonlab.blog/blog/best-open-source-pdf-t…	Secondary blog	E3
Unstract — why PDF-to-Markdown OCR fails for AI	unstract.com/blog/why-pdf-to-markdown-ocr-fa…	Vendor-authored	E3
PyMuPDF4LLM (GitHub)	github.com/pymupdf/pymupdf4llm	Project page	E3
77% of Employees Leak Data via ChatGPT (eSecurityPlanet)	esecurityplanet.com/news/shadow-ai-chatgpt-d…	News; reports vendor research	E2
What Data Are Employees Pasting Into ChatGPT? (CloudNuro)	cloudnuro.ai/blog/employee-chatgpt-data-leak…	Vendor blog	E2
Nearly 2 in 5 US workers… (Stacker)	stacker.com/stories/ai/nearly-2-5-us-workers…	Syndicated survey coverage	E2
Token Counters Compared (DevUtilHub)	devutilhub.pages.dev/blog/token-counters-com…	Secondary; tool vendor	E4
tokencalc vs OpenAI Tokenizer	tokencalc.pages.dev/compare/tokencalc-vs-ope…	Secondary; tool vendor	E4
EU AI Act Art. 50 labelling & metadata (sota.io)	sota.io/blog/eu-ai-act-art50-gpai-content-la…	Secondary; legal text not reviewed	E5
EU AI Act and C2PA (virtuall.pro)	virtuall.pro/blog/eu-ai-act-and-c2pa	Secondary	E5
C2PA Viewer — EU AI content labels	c2paviewer.com/articles/eu-ai-content-labels…	Secondary; tool vendor	E5, E6
C2PA & AI Metadata Remover (IPVoid)	ipvoid.com/c2pa-ai-metadata-remover/	Tool page	E6
Local PDF & AI image metadata remover (wavect)	wavect.io/tools/pdf-image-metadata-cleaner/	Tool page	E6
Transformers.js WebGPU in Next.js (noqta)	noqta.tn/en/tutorials/transformers-js-webgpu…	Tutorial	E7
Browser background removal with ONNX Runtime WebGPU (IMG.LY)	img.ly/blog/browser-background-removal-using…	Vendor blog	E7
Building a background remover with Transformers.js (LogRocket)	blog.logrocket.com/building-background-remov…	Tutorial	E7
JSON mode still fails: the recovery ladder (DEV)	dev.to/gabrielanhaia/json-mode-still-fails-h…	Community blog	E8

Source	Link	Type	Used for
Repair JSON with jsonrepair (AI SDK cookbook)	ai-sdk.dev/cookbook/node/repair-json-with-js…	Project documentation	E8
Measuring prompt injection in LLM resume screening (arXiv)	arxiv.org/html/2605.28999v1	Preprint; USENIX Security 2026 per coverage	E9
Detecting hidden prompt injections in PDFs (DEV)	dev.to/andy8647/i-built-a-tool-to-detect-hid…	Community blog	E9
Catching the wrong students — AI detection (HEPI)	hepi.ac.uk/2026/07/20/catching-the-wrong-stu…	Think-tank analysis	E10
Stanford TOEFL AI-detection study (summary)	workingeducators.org/blog/stanford-toefl-ai-…	Secondary summary; original 2023 paper not read	E10
Google AI Overviews impact on publishers (Search Engine Journal)	searchenginejournal.com/google-ai-overviews-…	Industry press	E11
Publisher traffic falls 33% (DesignRush)	news.designrush.com/ai-overviews-publisher-t…	Secondary	E11
Scaled content abuse (PPC Land)	ppc.land/scaled-content-abuse/	Secondary; Google's own policy page not reviewed	E12

Final product recommendation

> **GO / NO-GO RULE** Proceed to implementation after (1) a lightweight validation pass on the core categories and the four Wave-1 AI Prep candidates, (2) confirming that the core image and PDF workflows and the shared text workspace are reliable on target browsers and a lower-powered phone, and (3) deciding D1–D3 and the domain. Do not wait for perfect keyword data, and do not present unverified demand or competitor gaps as facts.

Launch with fewer reliable tools and genuinely useful pages rather than a catalogue of shallow utilities. The AI Prep category is a bet that preparing data for AI, and checking what comes back, is an under-served need — **the plan is built so that the bet is cheap to make and cheap to abandon.** Measure successful task completions first; use Search Console and real behaviour to decide what to expand.